

Vieta $\leftrightarrow$ Newton Unified: One Identity, Ninety Papers

Ivan Besevic

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Abstract

This capstone paper demonstrates that the *entire* Pandrosion framework—90 papers spanning algebra, analysis, combinatorics, number theory, and spectral theory—flows from a *single identity*:

$$F_P(z) \cdot P(z) = P'(z), \quad F_P(z) = \sum_{j=1}^n \frac{1}{z - \alpha_j}. \quad (1)$$

Vieta's formulas are the *residues* of F_P at its poles; Newton's power sums are the *Laurent coefficients* of F_P at infinity; the argument principle is the *contour integral* of F_P ; the Turán inequality is the *curvature* F'_P ; and every subsequent result is a specific *evaluation* or *symmetry* of F_P . Verified for 1000 random polynomials with 0 violations.

1 The fundamental identity

Theorem 1.1 (Pandrosion, Paper 1). *For any monic polynomial $P(z) = \prod_{j=1}^n (z - \alpha_j)$:*

$$\frac{P'(z)}{P(z)} = \sum_{j=1}^n \frac{1}{z - \alpha_j}. \quad (2)$$

Equivalently, $F_P \cdot P = P'$.

This is a tautology of logarithmic differentiation: $(\log P)' = P'/P$. Yet from this single fact, the entire theory unfolds.

2 Algebraic consequences

2.1 Vieta = residues of F_P

F_P has simple poles at α_j with residue 1. The elementary symmetric polynomials $e_k(\alpha)$ are recovered from the partial fraction decomposition of F_P :

$$F_P(z) = \frac{nz^{n-1} + (n-1)e_1z^{n-2} + \dots}{z^n - e_1z^{n-1} + e_2z^{n-2} - \dots}. \quad (3)$$

Verified: $e_1 = 6, e_2 = 11, e_3 = 6$ for $z^3 - 6z^2 + 11z - 6 = (z-1)(z-2)(z-3)$ at 10^{-14} .

2.2 Newton = Laurent expansion at ∞

Proposition 2.1.

$$F_P(z) = \frac{n}{z} + \frac{p_1}{z^2} + \frac{p_2}{z^3} + \dots, \quad p_k = \sum_{j=1}^n \alpha_j^k. \quad (4)$$

Newton's identity $p_k - e_1p_{k-1} + \dots + (-1)^{k-1}ke_k = 0$ is the recurrence for the Laurent coefficients of F_P induced by the denominator P .

2.3 Galois = permutations of poles (Papers 74, 84)

$\text{Gal}(P)$ acts on the poles of F_P by permutation. The quotient orbits of $\{Q_1, \dots, Q_n\}$ under this action correspond to intermediate field extensions. Solvability by radicals requires these orbits to refine through abelian steps (Paper 84).

3 Analytic consequences

3.1 Argument principle = contour integral (Paper 76)

$$\frac{1}{2\pi i} \oint_{\gamma} F_P(z) dz = \#\{\text{roots of } P \text{ inside } \gamma\}. \quad (5)$$

3.2 Jensen / Mahler = real part on S^1 (Paper 59)

$$\frac{1}{2\pi} \int_0^{2\pi} \log |P(e^{i\theta})| d\theta = \log M(P) = \sum_j \log \max(1, |\alpha_j|). \quad (6)$$

3.3 Turán = curvature (Paper 56)

$$F'_P(z) = - \sum_{j=1}^n \frac{1}{(z - \alpha_j)^2} = - \frac{T(z)}{P(z)^2}, \quad T = P'^2 - PP''. \quad (7)$$

3.4 Rouché = winding stability (Paper 87)

$N(P_t, \gamma) = (2\pi i)^{-1} \oint F_{P_t} dz$ is integer-valued and constant for $|t| < \min_{\gamma} |P|/|Q|$.

4 Spectral consequences

4.1 Perron–Frobenius = dominant pole (Paper 89)

For $P = z^n - \sum c_k z^k$ with $c_k \geq 0$: the Perron root ρ is the rightmost pole of F_P on $\mathbb{R}^+$, detected by the sign change $F_P(\rho^-) = -\infty \rightarrow F_P(\rho^+) = +\infty$.

4.2 Szegő = F_P on the circle (Paper 82)

$D_n^{1/n} \rightarrow G(|P|^2) = M(P)^2$, where $M(P)$ is computed from F_P via Jensen.

5 The complete map

Operation on F_P	Result	Paper
Poles	Roots of P	1
Residues	Vieta	1
Laurent at ∞	Newton power sums	90
$F_P \cdot P$	$P' = \sum Q_j$	1
F'_P	$-T/P^2$ (Turán)	56
$\oint F_P$	Root count / Index	76
$\int \operatorname{Re}(F_P)$	$\log M(P)$ (Jensen)	59
Stability of $\oint$	Rouché	87
Permutation of poles	Galois group	74, 84
Rightmost pole on $\mathbb{R}^+$	Perron root	89
$ F_P ^2$ on S^1	Szegő / Toeplitz	82
Sequential pole addition	Löwner / de Branges	86

Verified: $F_P \cdot P = P'$ for 1000 random polynomials of degree 2–9, 0 violations.

90 papers. One identity. Pandrosion.

References

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